

SIMATS ENGINEERING

NAME :Swetha.B

Reg no.: 192412215

ITA0606-

Machine Learning

Experiment -16

Compare different types Classification Algorithms and evaluate their performance

Code:

```
# Comparison of Classification Algorithms

import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder

from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import accuracy_score, classification_report

# Load dataset
df = pd.read_csv("play_tennis (1).csv")

# Convert categorical columns into numerical values
for column in df.select_dtypes(include=["object"]).columns:
    encoder = LabelEncoder()
    df[column] = encoder.fit_transform(df[column])

# Separate features and target
X = df.drop("play", axis=1)
y = df["play"]

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Feature scaling
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Create classification models
models = {
    "Logistic Regression": LogisticRegression(),
    "K-NN": KNeighborsClassifier(n_neighbors=5),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "Random Forest": RandomForestClassifier(
        n_estimators=100, random_state=42
    )
}

# Compare models
results = {}
```

```

print("Classification Algorithm Comparison")
print("-----")

for name, model in models.items():

    # Use scaled data for Logistic Regression and K-NN
    if name in ["Logistic Regression", "K-NN"]:
        model.fit(X_train_scaled, y_train)
        y_pred = model.predict(X_test_scaled)
    else:
        model.fit(X_train, y_train)
        y_pred = model.predict(X_test)

    accuracy = accuracy_score(y_test, y_pred)
    results[name] = accuracy

    print("\n", name)
    print("Accuracy: {:.2f}%".format(accuracy * 100))

# Display comparison
print("\n-----")
print("Performance Comparison")
print("-----")

for name, accuracy in results.items():
    print("{} : {:.2f}%".format(name, accuracy * 100))

# Find best algorithm
best_model = max(results, key=results.get)

print("\nBest Performing Algorithm:")
print(best_model)
print("Accuracy: {:.2f}%".format(results[best_model] * 100))

```

OUTPUT:

```

*** Classification Algorithm Comparison
-----

Logistic Regression
Accuracy: 66.67%

K-NN
Accuracy: 66.67%

Decision Tree
Accuracy: 100.00%

Random Forest
Accuracy: 66.67%

-----
Performance Comparison
-----
Logistic Regression : 66.67%
K-NN : 66.67%
Decision Tree : 100.00%
Random Forest : 66.67%

Best Performing Algorithm:
Decision Tree
Accuracy: 100.00%

```

